

Portfolio Milestone: Data Transformation and Data Model Design

This portfolio milestone focused on preparing the dataset for analysis using Power BI and Power Query Editor. The data source included the Orders, People, and Returns tables. First, the tables were imported into Power BI. The first row of each table was promoted to headers so the column names were recognized correctly. Next, data types were reviewed and assigned to the appropriate format, including text, dates, and numeric values. During the process, an extra Promoted Headers step caused an error in the Orders table because the headers had already been promoted. The unnecessary step was removed, leaving a single Promoted Headers step followed by the correct Changed Type step. The People and Returns tables were also verified to ensure the headers and data types were correct before selecting Close & Apply.

The data model was designed using Orders as the central fact table. The People table was connected to Orders through the Region field, allowing sales to be analyzed by regional manager. The Returns table was connected to Orders through the Order ID field to identify returned orders. The completed model provides a reliable foundation for dashboards and business analysis. Overall, the transformation process improved data quality and the relationships created an efficient and well-structured data model suitable for visualization and decision-making.

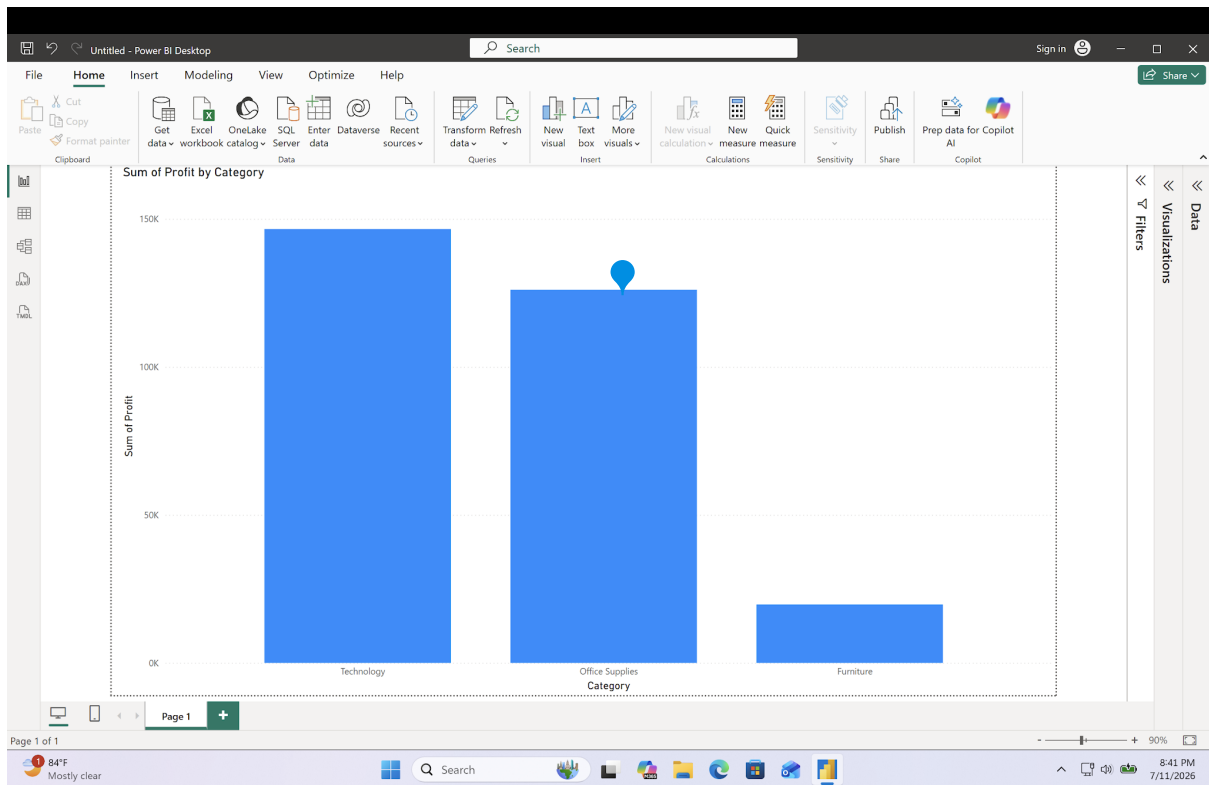
Screenshots

Power Query Editor interface showing a table with 21 columns and 999+ rows. The table contains data for orders, including Order ID, Order Date, Ship Date, Ship Mode, and Customer ID. The interface includes a ribbon with tabs like File, Home, Transform, Add Column, View, Tools, and Help. The bottom status bar shows the temperature as 84°F and the time as 8:35 PM on 7/11/2026.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID
1	US-2021-103800	1/3/2021	1/7/2021	Standard Class	DP-13000
2	US-2021-112326	1/4/2021	1/8/2021	Standard Class	PO-19195
3	US-2021-112326	1/4/2021	1/8/2021	Standard Class	PO-19195
4	US-2021-112326	1/4/2021	1/8/2021	Standard Class	PO-19195
5	US-2021-141817	1/5/2021	1/12/2021	Standard Class	MB-18085
6	US-2021-167199	1/6/2021	1/10/2021	Standard Class	ME-17320
7	US-2021-167199	1/6/2021	1/10/2021	Standard Class	ME-17320
8	US-2021-106054	1/6/2021	1/7/2021	First Class	JO-15145
9	US-2021-167199	1/6/2021	1/10/2021	Standard Class	ME-17320
10	US-2021-167199	1/6/2021	1/10/2021	Standard Class	ME-17320
11	US-2021-167199	1/6/2021	1/10/2021	Standard Class	ME-17320
12	US-2021-130813	1/6/2021	1/8/2021	Second Class	LS-17230
13	US-2021-167199	1/6/2021	1/10/2021	Standard Class	ME-17320
14	US-2021-167199	1/6/2021	1/10/2021	Standard Class	ME-17320
15	US-2021-105417	1/7/2021	1/12/2021	Standard Class	VS-21820
16	US-2021-105417	1/7/2021	1/12/2021	Standard Class	VS-21820
17	US-2021-135405	1/9/2021	1/13/2021	Standard Class	MS-17830
18	US-2021-135405	1/9/2021	1/13/2021	Standard Class	MS-17830
19	US-2021-149020	1/10/2021	1/15/2021	Standard Class	AI-10780
20	US-2021-149020	1/10/2021	1/15/2021	Standard Class	AI-10780
21	US-2021-130092	1/11/2021	1/14/2021	First Class	SV-20365
22	US-2021-157147	1/13/2021	1/18/2021	Standard Class	BD-11605
23	US-2021-109232	1/13/2021	1/16/2021	Second Class	ND-18370
24	US-2021-157147	1/13/2021	1/18/2021	Standard Class	BD-11605
25	US-2021-118192	1/13/2021	1/18/2021	Standard Class	MM-17920
26	US-2021-162775	1/13/2021	1/15/2021	Second Class	CS-12250
27	US-2021-162775	1/13/2021	1/15/2021	Second Class	CS-12250
28	US-2021-162775	1/13/2021	1/15/2021	Second Class	CS-12250
29	US-2021-118192	1/13/2021	1/18/2021	Standard Class	MM-17920
30	US-2021-162775	1/13/2021	1/15/2021	Second Class	CS-12250

Power Query Editor interface showing a table with 2 columns and 4 rows. The table contains data for regional managers, including Regional Manager and Region. The interface includes a ribbon with tabs like File, Home, Transform, Add Column, View, Tools, and Help. The bottom status bar shows the temperature as 84°F and the time as 8:35 PM on 7/11/2026.

Regional Manager	Region
Sadie Pawthorne	West
Chuck Magee	East
Roxanne Rodriguez	Central
Fred Suzuki	South



Reference

Microsoft. (n.d.). Power Query documentation.

<https://learn.microsoft.com/power-query/>